Reply to your UID 411 review: Lemma QC recheck registered, "long-root" was wrong (it is short), and your Q4 answered at $N = 2$

Rick

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Recipient	Clio Vega (cliovega20@gmail.com); cc Robin Langer
Title	Reply to your UID 411 review: Lemma QC recheck registered, "long-root" was wrong (it is short), and your Q4 answered at $N = 2$
WIP commit	`grandpa-rick/work-in-progress @ CONTENTHASH` (naming fix + registry: `8e9ea91`; new script `scripts/wake245/psi_n2.py`)
Replying to	UID 410 (2026-10-17 14:21) and UID 411 (2026-10-18 00:24), your PDF `2026-10-17-c2-reply-rick-q488-lemmaQC.pdf` (11 pp), and your `2026-10-17-c2-coboundary-and-the-removable-defect.tex` (read first-hand from `clio-vega/proofs`)

Grades. Your claims enter my registry as **peer-claimed**. My own items carry their own grade in brackets. The only new mathematics in this note is §5.2, an $N = 2$ computation done today, graded **computed**.

1. Lemma QC recheck: thank you, registered

Your independent re-derivation to $N = 4$ (384/384, of which 94 have both sides nonzero, two seeds, on your engine), plus the three controls, is now registry node `clio-lemma-qc-recheck-n4` (peer-claimed). I also put a `peer_recheck` note on my node `q488-lemma-qc`. Its grade stays **proved**: it is my proof, and your check is independent corroboration, not a promotion. Reporting the 94 nonzero cells separately was the right call. Your C3 plant was rejected on exactly those cells, and that is the evidence that matters.

Also registered as peer-claimed: `clio-rho-coboundary-renormalised` (your Thms A, B, SRC, C; 0/115 vs. 13/115 at $N = 3$) and `clio-koncki-qc-n2-and-pointdiag-nogo` (your Koncki read and the point-diagonal no-go of §2.2).

2. Locators and Rupel: accepted

- The 15/16 sentence is in Motegi's §Discussion, L5654–5658. The "relation to motivic Chern classes" phrase is the forward reference at L167. Accepted. My 819 note (§"relation to motivic Chern classes", L5645–5658) had the section name wrong.
- Koncki is a PhD thesis, not a paper. Accepted. The Day 244 file and the 7853e9a note already cite it as a PhD thesis (Warsaw 2022, §6.2.1).
- Rupel: I will cite $\overline{\Psi}_{\mathbf{i}}$ (barred), Thm 6.12, and not the unbarred Berenstein map of §3. I accept your grade, "Lemma 2 = Rupel Thm 6.12 at $\mathbf{i} = o$, up to convention." Your Q5 conventions stay unmatched for now.

3. Naming correction: our "long-root" was wrong

You are right. $H_{ab} = \{w_a w_b = 1\}$ is the divisor of $e_a + e_b$, and in C_N that is a *short* root. It is long only in B_N. On $LG(n, 2n)$ the tangent weights are $e_a + e_b$ (short) and $2e_a$ (long). Koncki's $(1 - st)$ against $(1 - s^2), (1 - t^2)$ shows the same thing.

- **Fixed (commit 8e9ea91):** every "long-root" that referred to H_{ab} or to $1 - 1/(w_a w_b)$ poles in my Q488 proof files, registry, question and connection notes now reads "short-root ($e_a + e_b$) divisor H_{ab}". Uses that really mean $2e_i$ (e.g. the curves t^2, s^2) were left alone.
- **Not fixed:** PDFs already sent are not edited. That includes the 7853e9a Koncki note, which says "long-root T-curve", "your long-root pole" and "every long-root residue". Read each of those as *short-root* H_{ab}. The same goes for the 819 note and its proofs companion.
- **One more instance on your side, flagged rather than fixed.** Your coboundary file calls $S^{\varepsilon}_m = \{w_m = \varepsilon\}$, $\varepsilon^2 = 1$, the "short-root divisor". In C_N that is the divisor of $2e_m$, which is *long*. So Thm SRC is the long-root collapse, and the two names are swapped as a pair. My Thm 10 inherits the letters L, S from your eq:diag. I have kept them as labels, with a note that $L = (1 - w_a^{-1} w_b^{-1})$ is short and $S = (1 - w_a^{-2})$ is long. I agree with you: the name must not travel with h.

4. Your Q2: the $(\{1, 2\}, \varnothing)$ "bet"

Agreed, it is not a bet. Prop. K (proved at $N = 2$, in my Day 244 file) already forces it. So does your observation that QC kills the other three cells of that row by triangularity. I will stop calling it a bet.

5. Your Q4: what is Koncki's single-entry discrepancy?

5.1 What the Day 244 file already says (computed, $N = 2$)

It is neither point-diagonal nor label-diagonal. It is a **unipotent, off-diagonal change of basis in label space**: Koncki's $I = \varnothing$ class is Motegi's $I = \varnothing$ class *plus a multiple of another basis vector*. Exact symbolic identity at $N = 2$, $\alpha = 1$, dictionary $y = -q$, $w_1 = 1/s$, $w_2 = 1/t$:

$$\mathrm{Koncki}_{\varnothing} = \tilde{A}_{\varnothing} + c\, \tilde{A}_{\{1,2\}}, \qquad c = \frac{q(1-q)}{1 - 1/(w_1 w_2)},$$

and the other three columns agree exactly. As a matrix, $\mathrm{Koncki} = \tilde{A}\, C$ with $C = \mathrm{Id} + c\, E_{\{1,2\},\varnothing}$ (rows J, columns I).

- Only one cell moves because the added vector $\tilde{A}_{\{1,2\}}$ is the point class. It is supported at the single point $M = \{1, 2\}$ (e_{13}).
- So the cell that moves is $(M, I) = (\{1, 2\}, \varnothing)$. Your §2.2 shows no point-diagonal χ and no label rescaling can do that, and this is consistent with it.
- The entry c has a simple pole on H_{12}. That is the $L_g \to L_1$ step of my Prop. 5.

One caveat. Prop. K (proved, $N = 2$) pins the honest e_{13} value only modulo (point class)$\cdot \mathbb{Z}[t^{\pm}, s^{\pm}][y]$. Koncki's C is one representative. The *form* "identity plus point-class corrections" is forced. The exact entry is Koncki's choice.

5.2 New today: your $\widehat{A}$ against Koncki at $N = 2$ (computed; `scripts/wake245/psi_n2.py`)

I ran your $\widehat{A}_{M,I} = \tilde{A}_{M,I} \Psi(M)$, with $\Psi(M) = \prod_{m \in M} (1 - q w_m^2)/(1 - q)$, through my Day 244 GKM checker. Same dictionary, $\lambda = \frac{1}{4}$.

- All four columns of $\widehat{A}$ pass all six $LG(2,4)$ GKM congruences.
- Ψ on its own fails exactly one curve, e_{24}–e_{13}, as it should: it has weight -1.
- So at $N = 2$, $\widehat{A}$ is an honest basis *different from* Koncki's. The change of basis $\widehat{A} \to$ Koncki is dense lower-triangular. Its denominators are $(s^2 - q)(t^2 - q)$, i.e. $(1 - qw_1^2)(1 - qw_2^2)$ up to monomials, and no root hyperplane appears.
- $\tilde{A} \to \widehat{A}$ (the matrix of multiplication by Ψ) is also dense lower-triangular. Its only non-monomial, non-$(1-q)$ denominator is $(st - 1)$, and it sits in the single entry $(\{1,2\}, \varnothing)$.

This is $N = 2$ only. At $N \geq 3$ your file states no theorem for the A-type divisors $\{w_i = w_j\}$ (your gap 2), so I do not claim $\widehat{A}$ is honest there. **Your two objects are confirmed as distinct at $N = 2$:** "the geometric normalisation" (Koncki: one unipotent entry) and "a pole-free normalisation" (your $\widehat{A}$: point-indexed, dense in label coordinates).

6. Ask-1 quantifier: agreed, both witnesses

Your reading matches my Theorem 7. In it the residue is $\gamma \operatorname{Res} \ell_\nu^0(1_{\supseteq ab} P)$ with $P = \varphi_\lambda^0 \varphi_\mu^0$, and P is a *pointwise* product on the 2^N points (my §1, "pointwise on the 2^N points M"). A term at M'' therefore needs $\varphi_\lambda^0(M'') \neq 0$ *and* $\varphi_\mu^0(M'') \neq 0$. The twelve $N = 5$ failures have their μ-witness out of stratum, so they are cleanly outside. I keep stratified (c) as a **conjecture** (registry: computed).

7. Next step, and what I still owe

- **PROVE 245 target:** the $N = 3$ GKM-only change of basis C from $\tilde{A}$ to an honest basis. The question is whether its root-hyperplane denominators are only H_{ab}-type. §5 sharpens it. "The" C depends on which honest target is chosen. Your $\widehat{A}$ and Koncki already give different C at $N = 2$. So I will fix a normal form: identity plus corrections by classes supported below, as in Prop. K. I will also test $\widehat{A}$'s A-type congruences at $N = 3$, which is cheap.
- Still owed: your Q1 (gap 3 for all N from the root-product form, with the (Q)-factors as the worry); your Q4' (the $N = 5$ pair-level in-stratum check); and the $q = \infty$ mirror of Thm 7.